

Course Code : ICE 3146 **Course Title :** Microprocessor and Interfacing Laboratory
Lab Task No. : 04
Lab Task Title/Topic : Study of Interfacing and Operating LEDs with the 8086 Microprocessor Kit in Both Kit and PC Modes.

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Title: Study of Interfacing and Operating LEDs with the 8086 Microprocessor Kit in Both Kit and PC Modes.

Abstract:

This experiment focuses on interfacing and operating Light Emitting Diodes (LEDs) with the 8086-microprocessor trainer kit in both Kit mode and PC mode. The main objective of this experiment is to understand how the 8086 microprocessor communicates with external hardware devices through input/output ports. In this experiment, LEDs are used as output devices to observe how the microprocessor sends signals to external components. Assembly language programs are written to control the LEDs, allowing them to turn on, turn off, or display different patterns. The experiment demonstrates the basic concept of hardware interfacing and helps students understand how microprocessors interact with peripheral devices. This practical work provides important knowledge about input/output operations, port addressing, and real-time control using the 8086-microprocessor.

Introduction (Theory):

The Intel 8086 is a 16-bit microprocessor that communicates with external devices through Input/Output (I/O) ports. Devices such as LEDs, switches, displays, and sensors are called peripheral devices, and connecting these devices with the microprocessor is known as interfacing. Through interfacing, the microprocessor can send data to control external hardware or receive data from input devices. In this experiment, LEDs are used as output devices to demonstrate how the microprocessor controls external hardware. An LED (Light Emitting Diode) is a semiconductor device that emits light when current flows through it. When the microprocessor sends binary data to an output port connected to LEDs, each bit of the data controls one LED. If the bit value is 1, the corresponding LED turns ON, and if the bit value is 0, the LED remains OFF. By changing the binary data pattern, different LED lighting sequences or patterns can be produced. The 8086 uses instructions such as OUT to send data from a register to an output port. The port address specifies which external device will receive the data. For example, if an 8-bit value is sent to an LED port, each bit controls one LED in the array. This allows the microprocessor to create different visual patterns by changing the data values in the program. The experiment can be performed in two operating modes of the microprocessor trainer kit: Kit Mode and PC Mode. In Kit Mode, machine codes are entered directly through the keyboard of the trainer kit and executed step by step. In PC Mode, the program is written on a computer using an assembler such as Microsoft Macro Assembler, and the generated machine code is then transferred and executed on the trainer kit. Through this experiment, students learn the basic concept of microprocessor I/O communication, hardware interfacing, and control of external devices. It also helps them understand how assembly language instructions are used to operate real hardware components like LEDs.

Material:

Hardware	Software
1. Intel 8086 Microprocessor Trainer Kit	1. Microsoft Macro Assembler (MASM)
2. LED Interface Module / LED Board	2. MDA-8086
3. Personal Computer (PC)	3. Text Editor (Notepad)
	4. Command Prompt (CMD)

Method:**Code:**

```
CODE SEGMENT
    ASSUME CS: CODE
    MOV AL, 10000000B
    OUT 1FH, AL
    MOV AL, 11111001B
    OUT 1BH, AL
    MOV AL, 11110110B
    OUT 1BH, AL
    MOV AL, 11111010B
    OUT 1BH, AL
    MOV AL, 11111111B
    OUT 1BH, AL
    INT 3
    CODE ENDS
END
```

Execution (Kit Mode):

1. First, the assembly language program was written in Notepad and saved as .asm file.
2. Machine code execution using CMD-

C:\MDA\8086\Asm8086>MASM.exe SUM26A13.asm

Object filename [SUM26_1.OBJ]: SUM26A13

Source listing [NUL.LST]: SUM26A13

Cross-reference [NUL.CRF]:

0 Warning Errors

0 Severe Errors

3. The 8086-microprocessor trainer kit was turned ON and reset.
4. The data entry mode was selected by pressing the DA key.
5. The Machine Code was entered manually using the Number Pad from the Kit.
6. The + key was pressed to move to the next memory location.
7. After entering all machine codes, the REG key was pressed to check the register values.
8. The program was executed step by step by pressing the STP key. After each STP key press,

the LEDs turned on and off according the code.

Execution (PC Mode):

1. First, the assembly language program was written in Notepad and saved as .asm file.
2. Machine code execution using CMD-

```
C:\MDA\8086\Asm8086>MASM.exe SUM26A13.asm
```

Object filename [SUM26_1.OBJ]: SUM26A13

Source listing [NUL.LST]: SUM26A13

Cross-reference [NUL.CRF]:

0 Warning Errors

0 Severe Errors

9. The MDA (Microprocessor Development and Analysis) software was opened.
10. The SUM26A13.asm file was loaded using the CMD panel (LO).
11. The program was assembled and linked to generate the .ABS (Absolute) file using the Assemble & Link option.
12. The generated .ABS file was loaded into the simulator.
13. The program was executed using the Run command.
14. The output and register values were observed through the simulator window and LED panel.

Result:

Result for Kit Mode:

		
Initial condition of LED	LED output for 11111001B	LED output for 11110110B
		
LED output for 11111010B	LED output for 11111111B	

Fig 1: LED outputs for different Sequence of Code.

Result for PC Mode:



Fig 2: Program execution in PC Mode using 8086 monitor terminal.

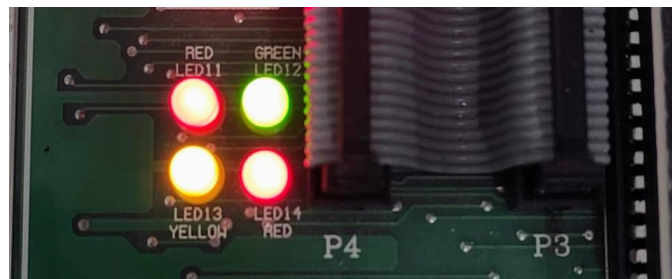


Fig 3: LED output for PC Mode.

Discussion:

This experiment demonstrated the interfacing and operation of LEDs with the 8086 microprocessor in both Kit Mode and PC Mode. Different binary data were sent to the output port using the OUT instruction, and the LEDs responded according to the transmitted patterns. In Kit Mode, the program was executed manually through the trainer kit, while in PC Mode it was assembled and executed using MASM and MDA-8086 software. The observed LED outputs matched the programmed data patterns in both modes. This experiment helped us understand I/O port communication, hardware interfacing, and the use of assembly language for controlling external devices with the 8086 microprocessor.